

Question 1 : Describe different types of data sources used in ETL with suitable examples.

Answer:-

In an **ETL (Extract, Transform, Load)** process, data is collected from different **data sources**, transformed into a suitable format, and then loaded into a target system like a data warehouse. These data sources can be categorized based on their structure, origin, and format.

Types of Data Sources in ETL (with Examples)

1. Structured Data Sources

These sources store data in a **well-defined schema** (rows and columns).

- **Examples:**
 - Relational databases (MySQL, Oracle, SQL Server)
 - ERP systems (SAP, Oracle ERP)
 - CRM systems (Salesforce)
 - **Characteristics:**
 - Easy to query using SQL
 - Highly organized
 - Consistent data format
 - **Example Use Case:**

Extracting customer records from a **MySQL database** for reporting.
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2. Semi-Structured Data Sources

These sources do not follow a strict table structure but still contain **organizational tags or markers**.

- **Examples:**
 - JSON files
 - XML documents
 - NoSQL databases (MongoDB, CouchDB)
- **Characteristics:**
 - Flexible schema
 - Data organized using tags or key-value pairs

- **Example Use Case:**
Extracting API response data in **JSON format** for analytics.
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3. Unstructured Data Sources

These contain data that has **no predefined format or structure**.

- **Examples:**
 - Text files
 - Emails
 - Images, audio, video
 - Social media posts
 - **Characteristics:**
 - Difficult to process directly
 - Requires preprocessing (e.g., NLP, image recognition)
 - **Example Use Case:**
Extracting sentiment from **Twitter posts** for market analysis.
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4. Internal Data Sources

Data generated **within an organization**.

- **Examples:**
 - Transaction systems
 - HR systems
 - Inventory databases
 - **Characteristics:**
 - Reliable and controlled
 - Easily accessible
 - **Example Use Case:**
Using **sales transaction data** from an internal billing system.
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5. External Data Sources

Data collected from **outside the organization**.

- **Examples:**
 - Third-party APIs (weather, stock market)

- Government datasets
 - Vendor data feeds
 - **Characteristics:**
 - May have quality or consistency issues
 - Requires validation
 - **Example Use Case:**

Integrating **weather data API** into a logistics system.
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6. Batch Data Sources

Data that is collected and processed in **large chunks at scheduled intervals**.

- **Examples:**
 - Daily sales reports
 - Monthly financial files
 - **Characteristics:**
 - High volume
 - Not real-time
 - **Example Use Case:**

Loading **daily transaction logs** into a data warehouse overnight.
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7. Real-Time (Streaming) Data Sources

Data that is generated and processed **continuously in real time**.

- **Examples:**
 - IoT sensors
 - Website clickstreams
 - Stock trading systems
 - **Characteristics:**
 - Requires fast processing
 - Low latency
 - **Example Use Case:**

Processing **live user activity** on an e-commerce site.
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8. Flat Files

Simple file-based data sources without complex relationships.

- **Examples:**
 - CSV files
 - Excel spreadsheets
 - Text files
- **Characteristics:**
 - Easy to store and transfer
 - Limited structure
- **Example Use Case:**

Importing **CSV sales data** into a reporting system.

Question 2: What is data extraction? Explain its role in the ETL pipeline.

Answer:-

Data extraction is the process of collecting data from different sources for further processing.

Role in ETL:

- First step of ETL process
- Collects data from multiple sources
- Prepares data for transformation
- Ensures initial data quality
- Helps integrate data from different systems

👉 Without extraction, ETL cannot proceed.

Question 3: Explain the difference between CSV and Excel in terms of extraction and ETL usage.

Answer:-

CSV is a simple, text-based format that stores only raw data in rows and columns, making it easy to extract and ideal for ETL processes due to its simplicity and fast processing.

Excel is a complex, semi-structured format that includes multiple sheets, formulas, and formatting, making data extraction more difficult and requiring additional transformation in ETL.

Feature	CSV File	Excel File
Format	Plain text	Binary/structured format
Complexity	Simple	Complex (multiple sheets)
Speed	Faster processing	Slower
Storage	Lightweight	Larger size
ETL Usage	Easy to extract	Requires parsing tools
Features	No formatting	Supports formulas, charts

👉 CSV is preferred in ETL due to simplicity and speed.

Question 4: Explain the steps involved in extracting data from a relational database.

Answer:-

Steps involved in extracting data from a relational database

1. Identify Data Requirements

First, clearly define what data is needed for analysis or ETL. This includes identifying business needs, required fields, and time ranges.

2. Understand Database Schema

Study the database structure (tables, relationships, keys). Identify primary keys, foreign keys, and how tables are linked.

3. Establish Database Connection

Connect to the database using tools or drivers such as JDBC, ODBC, or database clients. Proper authentication (username/password) is required.

4. Select Relevant Tables and Columns

Choose only the necessary tables and columns to avoid unnecessary data extraction and improve performance.

5. Write SQL Queries

Create SQL queries using:

- **SELECT** to retrieve data
- **JOIN** to combine multiple tables
- **WHERE** to filter records
- **GROUP BY / ORDER BY** if needed

6. Apply Filters and Conditions

Use conditions (e.g., date ranges, specific values) to extract only relevant data, reducing data size and improving efficiency.

7. Execute the Query

Run the SQL query on the database server to fetch the data.

8. Handle Data Types and Data Quality

Check for null values, duplicates, and ensure correct data types (e.g., date, numeric). Perform basic cleaning if required.

9. Extract Data in Required Format

Convert or export the data into formats like CSV, Excel, or load it directly into a staging area or data warehouse.

10. Validate Extracted Data

Verify the extracted data for accuracy, completeness, and consistency with the source database.

👉 These steps ensure that data extraction is accurate, efficient, and ready for further transformation in the ETL process.

Question 5: Explain three common challenges faced during data extraction.

Answer:-

1. Data Quality Issues

- Missing or incorrect data
- Duplicate records

2. Data Integration Problems

- Different formats and structures
- Difficult to combine data

3. Performance Issues

- Large data volume slows extraction
 - Network or system limitations
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Question 6: What are APIs? Explain how APIs help in real-time data extraction.

Answer:-

APIs stand for Application Programming Interfaces. An API is a tool or medium that allows two software systems to communicate with each other (allow communication between different systems).

Role in Real-Time Data Extraction:

- Provide live data access
- Enable automatic data fetching
- Support continuous data flow
- Used in apps like weather, stock market

 **Example:** Fetching live weather data using API.

Question 7: Why are databases preferred for enterprise-level data extraction?

Answer:-

Reasons:

- High data reliability

- Fast query processing
- Handles large volumes of data
- Supports security and access control
- Ensures data consistency

👉 Databases are scalable and suitable for large organizations.

Question 8: What steps should an ETL developer take when extracting data from large CSV files (1GB+)?

Answer:-

Steps:

1. Use **chunk processing** (read data in parts)
2. Avoid loading entire file into memory
3. Use efficient tools (Python, Pandas, Spark)
4. Perform filtering during extraction
5. Optimize system memory usage
6. Validate data while processing
7. Store intermediate results if needed

👉 This improves performance and prevents system crashes.